Vikash Mishra

me24mtech12008@iith.ac.in | vikashmishra858@gmail.com | +91 9110015189

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.73
The Aryan International School	Class XII (CBSE)	2015	90.20%
Urmil Unique Central Academy	Class X (CBSE)	2013	95.00%

PROFESSIONAL SUMMARY

- Results-driven Executive Design Engineer with 3+ years of industry experience, currently pursuing M.Tech in Mechanics & Design at IIT Hyderabad (Dept. of Mechanical & Aerospace Engineering) and Selected in JST LOTUS 2026 Programme for advanced research in MEMS-based Inertial sensor design and fabrication at Kyoto University, Japan.
- Strong expertise in advanced mechanical design, modeling, and analysis, with hands-on experience in product development and engineering supervision.
- Proficient in SolidWorks, CATIA, PTC Creo, DraftSight, and ME-10, with working exposure to CoventorMP MEMS+, CoventorWare, KLayout, COMSOL Multiphysics and semiconductor packaging technologies.
- Solid foundation in GD&T, ECN/ECR processes, ETO environments, and VAVE methodologies for cost-effective and manufacturable designs.
- Seeking a challenging design/R&D role to leverage combined industrial experience and academic research in precision engineering, MEMS, and microsystem technologies.

SCHOLASTIC ACHIEVEMENTS

JST LOTUS 2026 - Selected Researcher

Selected for the prestigious JST LOTUS 2026 (India-Japan Circulation of Talented Youths in Science Programme) through a competitive international selection process for advanced research in MEMS-based Inertial sensor design and fabrication at Kyoto University, Japan.

JICA-JST 2.0 Scholarship 2026

Awarded JICA-JST funded research support, facilitating advanced research, technical training and scientific collaboration with Japan.

MEMS Inertial Sensors Summer School 2026

Selected participant at the MEMS Inertial Sensors Summer School 2026, Tohoku University, Japan, gaining advanced exposure to MEMS fabrication, accelerometer and gyroscope design.

INDUSTRIAL / PROFESSIONAL EXPERIENCE

Engineering Supervisor / Design Engineer II-- Regal Rexnord Corporation

- Managed the work of six individuals as an engineering supervisor, including an intern, two Graduate Engineer Trainees (GET), and three new engineers.
- Good at designing, unique Non-standard ETO Cases, with personalized input from customers.
- Made reports and Standard Operating Procedures (SOPs) for new product lines and assist in quality checking.
- Provided training to teams and others on topics like ISO balancing standards, Tolerance grades, and CAD tools.

Design Engineer I -- Rexnord Corporation, Hyderabad

- Specialized training in designing standard ETO cases, with a focus on XTSR, SERIES, and CMR/AMR Couplings.
- Worked on non-standard categories, particularly Locking Assembly Devices.

Engineering Intern -- Rexnord Corporation, Hyderabad

- Gained Product Knowledge and corporate behavior, and trained in designing standard ETO cases.

Summer Intern -- South Central Railway, Secunderabad

- Performed buffer-height assessment for ICF coaches.
- Developed calculations for determining buffer height under varying operating parameters.

SKILLS

CAD & Design Tools:

SolidWorks, PTC Creo, DraftSight, ME-10, CATIA, 3D Modelling, MS Office, Power BI, Photoshop

Design & Manufacturing Engineering:

GD&T, Tolerance Analysis, Product Standardization, Quality Inspection, SOP Development, Six Sigma, Lean Manufacturing, Kaizen, Design for Manufacturability.

CAE & Multiphysics Simulation:

ANSYS, Abaqus, COMSOL Multiphysics, CoventorWare, MEMS+, MATLAB/Simulink, FEM, MEMS Design and Simulation, Dynamic Analysis

Control & Signal Processing:

PLL, AGC, Closed-Loop Control, Resonance Tracking, Frequency Sweep, Vibration Analysis, Noise Measurement, Acoustic Characterization, FRF Analysis, SPL Measurement

RELEVANT COURSES

Mechanical Engineering: Engineering Mechanics, Strength of Materials, Theory of Machines,Machine Design,Mechanical Vibrations, Fluid Mechanics, Heat Transfer, Manufacturing Processes, Materials Engineering, CAD / Computer-Aided Engineering

Computational / Mathematical: Linear Algebra, Differential Equations, Probability & Statistics, Numerical Analysis, MATLAB, Finite Element Analysis

MEMS Engineering: MEMS & Microsystems, Sensor Technology, Microfabrication, Semiconductor Packaging, Control Engineering

POSITIONS OF RESPONSIBILITY

As Engineering Supervisor / Design Engineer II - Regal Rexnord

1. Led and coordinated a six-member engineering team comprising an intern, two Graduate Engineer Trainees and three engineers, managing design activities, technical support and work allocation.
2. Prepared SOPs, engineering reports and product documentation for new product lines and supported quality-review activities.

Head - Point of Contact (POC), Mirai Setu | India-Japan Collaboration, Kyoto, Japan

- Serving as Head POC for Mirai Setu, an India-Japan collaborative programme scheduled for October 2026 in Kyoto, Japan, coordinating communication and engagement between participating Indian and Japanese Delegates.

PROJECTS

Noise & Vibration Characterization and Control of FDM 3D Printer

- Investigated acoustic and vibrational behavior of an FDM 3D printer through experimental noise measurement.
- Applied accelerometer-based FRF analysis, microphone-based SPL evaluation, and MATLAB-driven spectral analysis.
- Designed and experimentally validated a passive vibration isolation system, significantly reducing resonance amplification.

3-RRR Parallel Planar Mechanism

- Designed and implemented a 3-RRR parallel planar robotic mechanism for tracking end-effector motion along

specified trajectories.
- Developed transformation-based motion control using Arduino Uno and servo motors.
- Analysed kinematic transformation and coordinated actuator motion for planar robotic operation.

Lever-Based Sensitivity Amplification in Single Proof-Mass MEMS Gyroscope

- Designed and analysed a lever-based displacement amplification mechanism for a single proof-mass MEMS gyroscope to enhance mechanical displacement and sensing sensitivity.
- Developed and simulated the complete MEMS gyroscope architecture using MEMS+, CoventorWare and MATLAB/Simulink, analysing dynamic behaviour, mode shapes and frequency response.
- Implemented open-loop and closed-loop drive-mode control, incorporating Automatic Gain Control (AGC) and Phase-Locked Loop (PLL) for stable oscillation amplitude and resonance tracking.
- Investigated electromechanical dynamics and system-level control architecture for improved gyroscope operation.

Closed-Loop Modeling and Simulation of MEMS-TMR Magnetometer

- Developed closed-loop simulation framework for a MEMS-TMR magnetometer, integrating mechanical resonator dynamics, spintronic sensing, and low-noise readout electronics using MATLAB/Simulink.
- Performed frequency-sweep and sensitivity analysis, including absolute and normalized sensitivity extraction, modulation depth estimation, and resistance variation characterization under varying DC magnetic fields.

Hardware Validation of Closed-Loop Quadrature Error Cancellation (QEC) for a Dual-Mass MEMS Gyro

- Developed and experimentally validating a closed-loop QEC architecture for a Dual-Mass MEMS Gyroscope (DMG) integrating PLL, AGC and QEC control loops to suppress quadrature error and improve gyroscope bias stability.
- Designed and analysed the complete MEMS gyroscope system, including reduced-order modelling, comprehensive noise modelling, control-loop stability/robustness analysis.
- Fabricating, packaging and characterizing DMG devices, followed by integration with a low-noise analog front-end and FPGA/SoC-based real-time control platform for digital demodulation, PLL, AGC and QEC implementation.

MEMS Bi-Stable Advance Mechanical Switch Design

- Designing a miniaturized bi-stable MEMS mechanical switch integrating a single electro-thermal compliant actuator with a curved-beam bistable mechanism to achieve electrically detectable two-state mechanical switching.
- Optimizing the coupled actuator bistable mechanism through thermal-mechanical analysis.
- Developing fabrication-ready MEMS mask layouts based on SOI/DRIE process constraints, including etch-hole optimization.

INDUSTRIAL ACHIEVEMENTS

Achievements

- Achieved 95% average On-Time Delivery (OTD) and 98% average First Pass Yield (FPY) during industrial tenure.
- Reviewed and provided technical comments on 20+ 8D quality reports.
- Completed Level-2 Laboratory Training in Semiconductor Packaging organised by CMTI Bengaluru.
- Journal paper "Closed loop control strategies for lever-based amplification of MEMS gyroscope" accepted for publication in ISSS, Springer Nature.

EXTRACURRICULAR

Extracurricular Activities

- Participate in competitive Weight Lifting and Powerlifting in inter college and university level tournaments.
- Volunteered in the Inter-IIT Sports Meet 2025 hosted at IIT Hyderabad, contributing to event coordination, participant support and smooth execution of inter-institute sporting activities.
- Volunteered and participated in APSCON 2025 at IIT Hyderabad, supporting conference activities while engaging with the academic and research community.